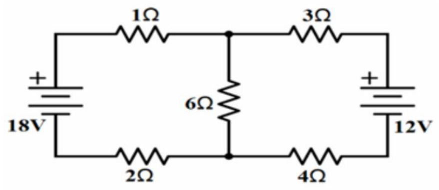
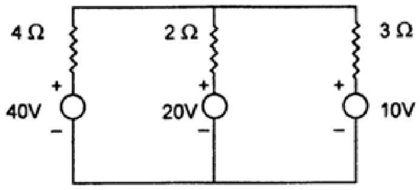
Practice Questions on Mesh Analysis

Q:4. Calculate the current in 6Ω resistor in the following circuit using mesh analysis. [2023-24]



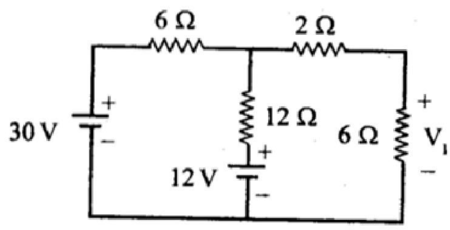
[Ans: 2 A]

Q:5. Find current in 2 Ω resistance in the following figure using loop analysis method. [2015-16]



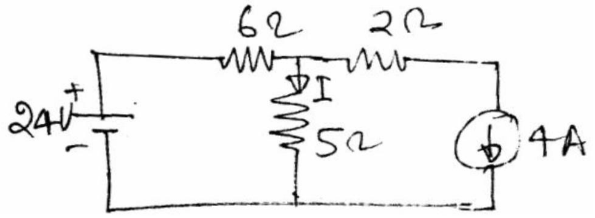
[Ans: 0.77 A]

Q:6. Find the voltage V_1 across 6 Ω resistance in the following circuit using loop analysis method. [2009-10, 2013-14]



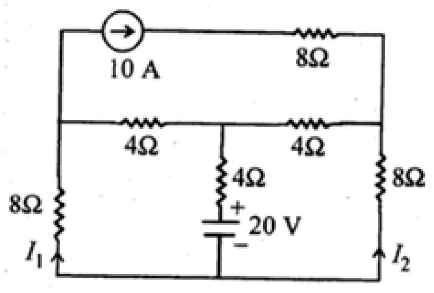
[Ans: 12 V]

Q:7. Find the current I in the circuit shown in figure below. [2006-07]



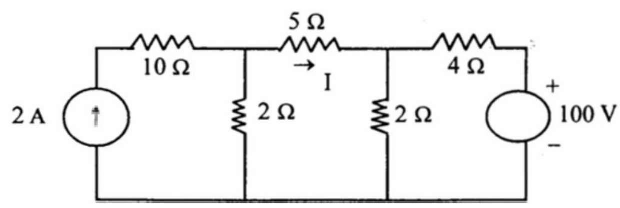
[Ans: 0 A]

Q:8. Using Mesh analysis, calculate the currents I_1 and I_2 in the fig. [2011-12]



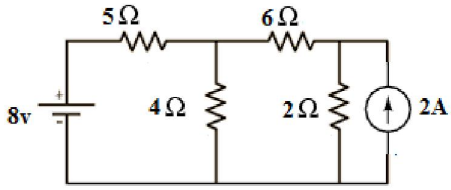
[Ans: 2.33 A, -4.33 A]

Q:9. Apply Mesh analysis, obtain the current through 5 ohm resistance in the following circuit. [2011-12, 2013-14]



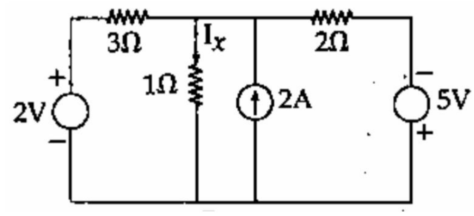
[Ans: -3.52 A]

Q:10. Find current in 4 Ω resistor in figure using Mesh analysis. [2023-24]



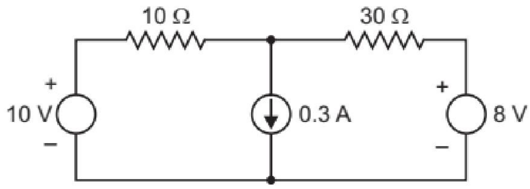
[Ans: 0.91 A]

Q:11. Using Mesh analysis method determine the current Ix in the following circuit. [2005-06]



[Ans: 0.091 A]

Q:12. Use mesh current method to determine currents through each of the resistors in the circuit shown in figure.



[Ans: $I_{10\Omega}$=0.275 A, $I_{30\Omega}$=-0.025 A]

6. Nodal Analysis:

Nodal analysis is based on Kirchhoff's current law.

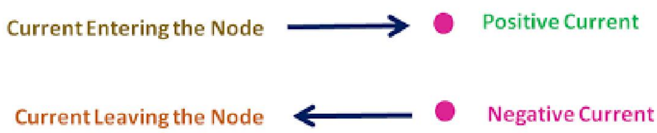
Let us consider a simple dc network as shown in figure below to find the currents through different branches using **"Nodal analysis"** method.

Step-I: *Inspect the total no. of node* in the circuit

Step-II: *Select one node as the reference node* (zero potential) and *label the remaining non-reference nodes as unknown node voltages* a $V_1, V_2, V_3 \dots\dots V_n$.

Step-III: *Assign branch currents in each branch* as $I_1, I_2, I_3 \dots\dots I_m$.(The choice of direction is arbitrary).

Step-IV: *For each non-reference node, apply KCL*: The sum of currents leaving (or entering) a node is equal to zero. Express each current in terms of the node voltages using Ohm's Law.



Step-V: *Express the branch currents in terms of node voltages.*

Now, Current in each branch can be determine as,

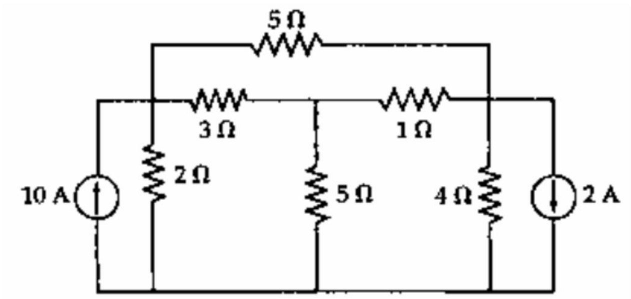
$$I_1 = \frac{V_1}{2} = \frac{-7.33}{2} = -3.665\ A$$

$$I_2 = \frac{V_1 - V_2}{10} = \frac{-7.33 + 5.33}{10} = -0.2\ A$$

$$I_3 = \frac{V_2}{4} = \frac{-5.33}{4} = -1.33\ A$$

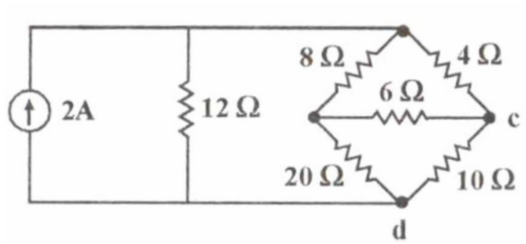
Practice Questions on Nodal Analysis

Q:16. Use nodal analysis to find the currents in various resistors of the circuit shown. [2005-06]



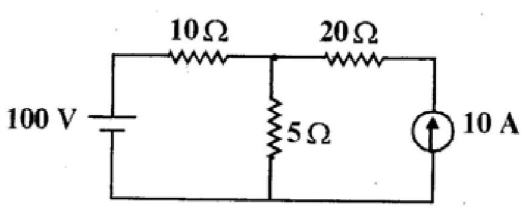
[Ans: $I_{2\Omega}=10A$, $I_{3\Omega}=7.714A$, $I_{5\Omega(vertical)}=2.286A$, $I_{1\Omega}=0.286A$, $I_{4\Omega}=2A$, $I_{5\Omega(top)}=2.286A$]

Q:17. Using Nodal analysis, find V_{cd} for the circuit shown in fig. [2007-08]



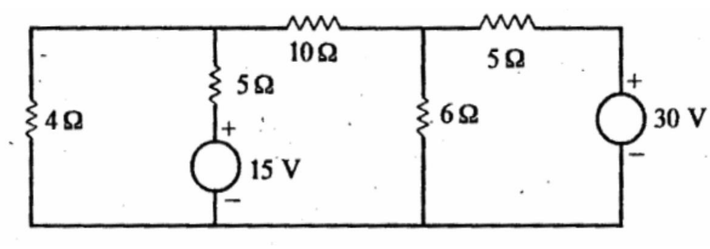
[Ans: 0.714 V]

Q:18. Find the current in all the resistive branches of the circuit shown below by (i) KVL (ii) KCL. [2007-08]



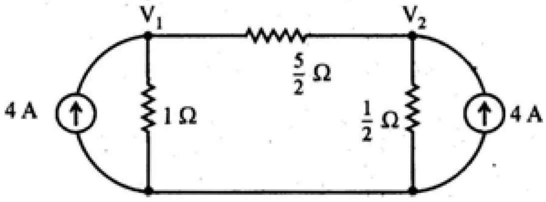
[Ans: $I_{10\Omega}=10$ A, $I_{20\Omega}=10$ A, $I_{10\Omega}=0$ A]

Q:19. Using Nodal analysis, find the current through 10 Ω resistor shown in fig. [2011-12]



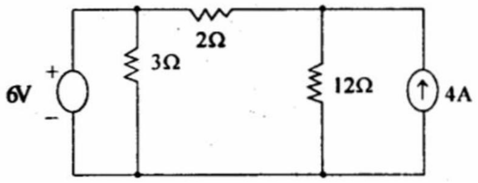
[Ans: 0.65 A]

Q:20. Determine the voltages V_1 and V_2 in the fig. [2011-12]



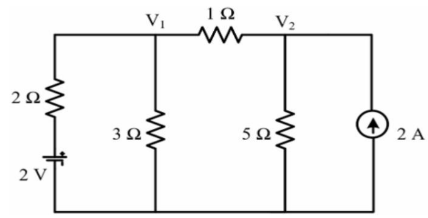
[Ans: V_1=3.5 V, V_2=2.25 V]

Q:21. Calculate currents in all the resistances of the circuit using node analysis method. [2013-14]



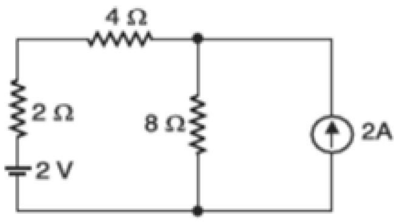
[Ans: $I_{3\Omega}$=0.33 A, $I_{2\Omega}$=1 A, $I_{12\Omega}$=0.25 A]

Q:22. Using Nodal analysis, find the current through 1Ω resistance shown in fig. [2016-17]



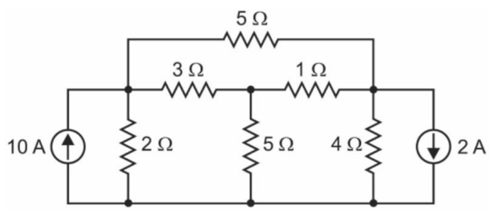
[Ans: 1.22 A]

Q:23. Using nodal analysis to find the voltage across and current through 4Ω resistor in figure. [2021-22]

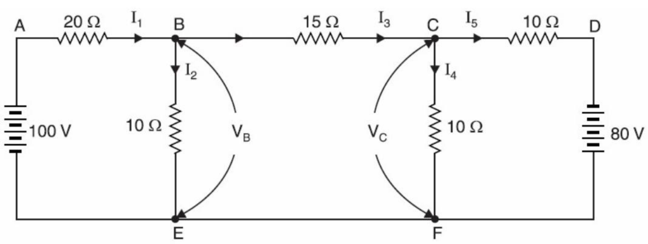


[Ans: -1 A, -4 V]

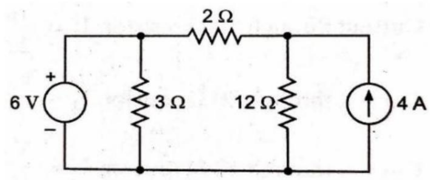
Q:24. Use nodal analysis to find the currents in various resistors of the circuit shown below. [2022-23] $[Ans{:}\ I_{1\Omega} = 1.3\ A, I_{2\Omega} = 6.025\ A, I_{3\Omega} = 2.32\ A, I_{4\Omega} = 0.95\ A, I_{5\Omega} = 1.02\ A, I_{5\Omega} = 1.65\ A]$



Q:25. Determine the currents in the various branches of the circuit shown below by using nodal analysis. [2022-23] $[Ans{:}\ I_1 = 6.41\ A, I_2 = -2.82\ A, I_3 = 0.59\ A, I_4 = -3.71\ A, I_5 = 4.29\ A]$

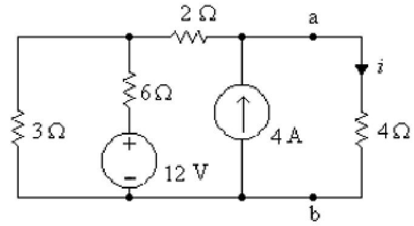


Q:26. Using nodal analysis find the branch currents in the circuit. (2023-24)



[Ans: $I_{3\Omega} = 2\ A, I_{2\Omega} = 3\ A, I_{12\Omega} = 1\ A$]

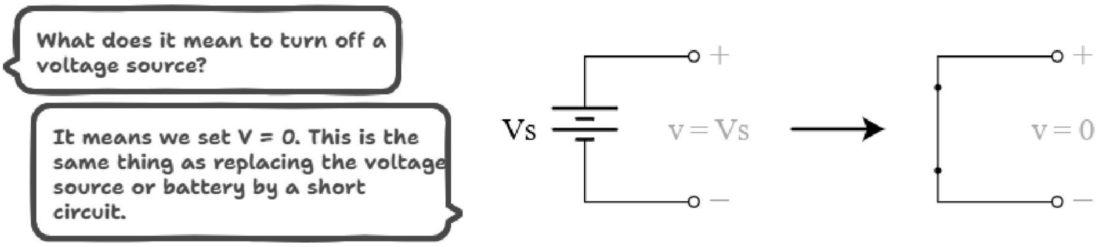
Q:27. Find current i using Nodal analysis in given circuit. [2023-24]

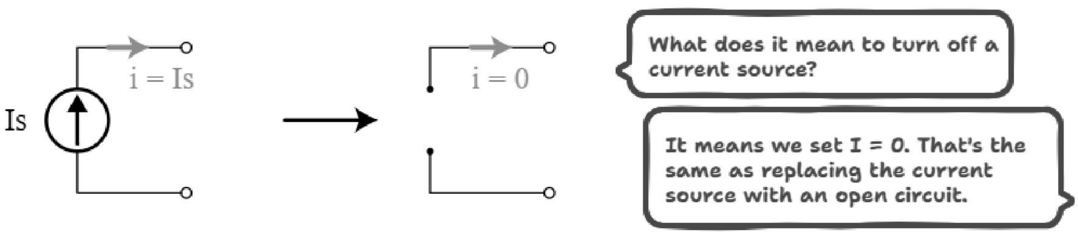


[Ans: $i = 2.5\ A$]

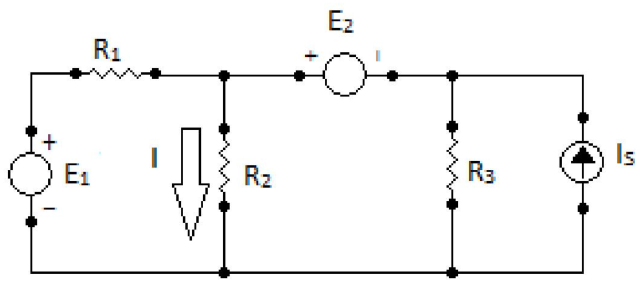
7. Superposition theorem: According to superposition theorem,

"In any linear bilateral network containing two or more independent sources, the net response in any branch is the algebraic sum of responses caused by each independent source acting alone, with all other independent sources being turned off."



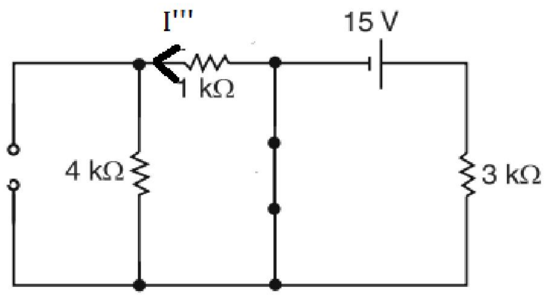


Consider a simple resistive network as shown in figure below. It has two independent practical voltage sources and one practical current source.



- **Case 3: When voltage source of 15 V is connected alone.**

The current through 1kΩ resistor due to 15V source acting alone is found by replacing the 10mA current source by an open circuit and 25 V source by a short circuit as shown in fig.



Here the resistor of 4Ω & 1Ω becomes shorted. The short circuit prevents any current from flowing in the 1 k Ω resistor.

Hence,

$$I''' = 0\ A$$

☞ Now by using superposition theorem, net current in 1 kΩ due to all the source,

$$I = I' - I'' + I'''$$
$$I = (8 \times 10^{-3}) - (5 \times 10^{-3}) + 0$$
$$I = 3 \times 10^{-3}\ A$$
$$\boldsymbol{I = 3\ mA}$$

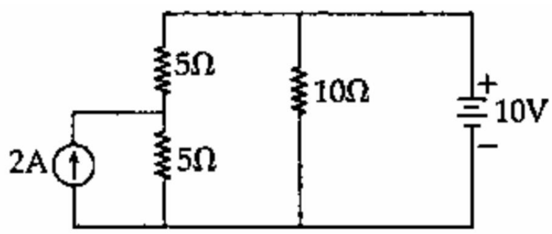
Limitations of superposition Theorem: -

1: Superposition theorem doesn't work for power calculation. Because power calculations involve either the product of voltage and current, the square of current or the square of the voltage, they are not linear operations.

2: Superposition theorem cannot be applied for non-linear circuit

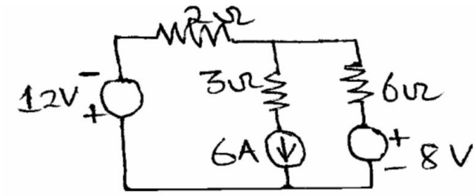
Practice Questions on Superposition Theorem

Q:29. Using superposition theorem, determine currents in all the resistances of the following network: [2005-06, 2013-14]



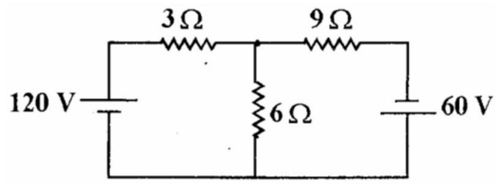
[Ans: $I_{5\Omega}$=2 A, $I_{5\Omega}$=0 A, $I_{10\Omega}$=1 A]

Q:30. Determine current in all branches of the circuit shown in fig. using Superposition theorem. [2006-07]



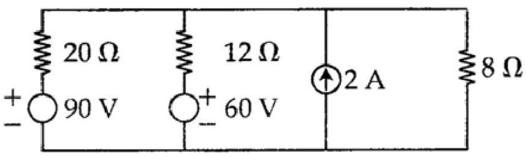
[Ans: $I_{2\Omega}$=2 A, $I_{3\Omega}$=6 A, $I_{6\Omega}$=4 A]

Q:31. In the circuit shown in fig, find the current through the 6Ω resistor using superposition theorem. [2007-08]



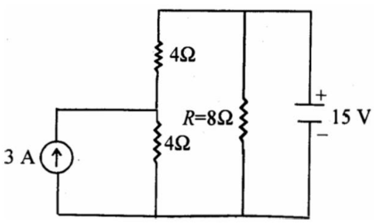
[Ans: 9.1 A]

Q:32. Using superposition theorem find the current in 8 Ω resistor of the circuit shown in figure. [2009-10]



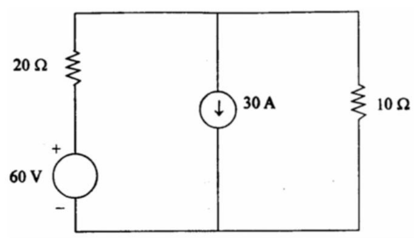
[Ans: 5.57 A]

Q:33. Using superposition theorem, find the current flowing through the resistor R in the fig. [2011-12]



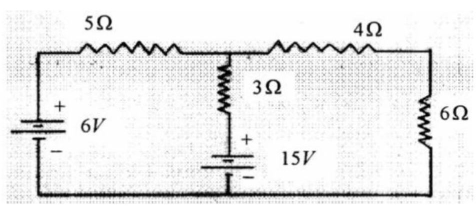
[Ans: 1.875 A]

Q:34. Find the current flowing through 10 Ω resistance in the following circuit. Use superposition theorem. [2011-12, 2013-14]



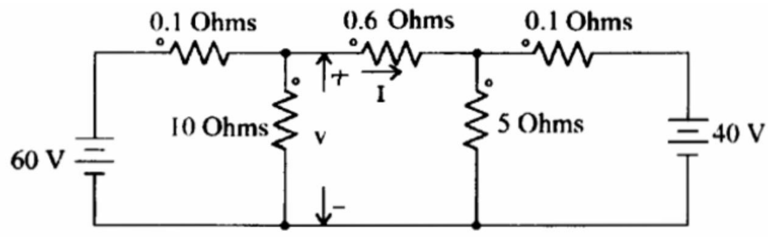
[Ans: 18 A]

Q:35. State and explain superposition theorem. Determine the current through 6 Ω resistor. [2014-15]



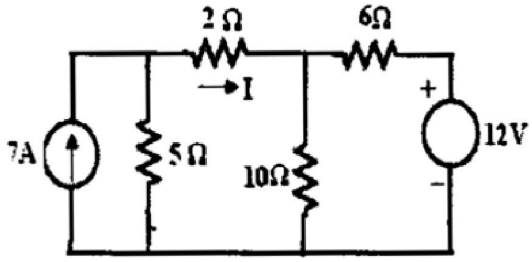
[Ans: 0.979 A]

Q:36. Find the V and I in the given circuit using superposition theorem. [2014-15]



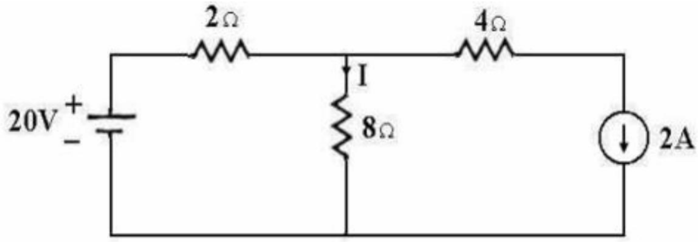
[Ans: 56.871 V, 26.52 A]

Q:37. Find current through 2 Ω resistance using superposition theorem. [2018-19]



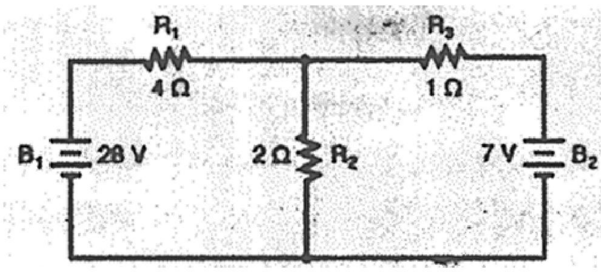
[Ans: 2.9 A]

Q:38. Determine current through 8 Ω resistor and power in the 4Ω resistor in the network shown in fig. using superposition theorem. [2018-19]



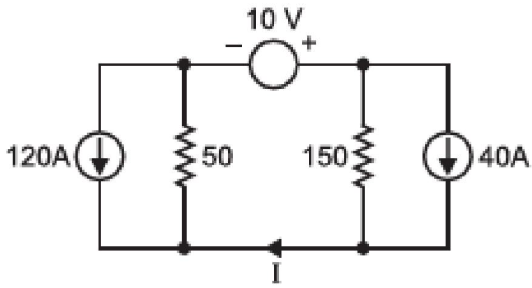
[Ans: 1.6 A, 16 W]

Q:39. Using superposition theorem, find the current flowing through 2 Ω resistance in the following circuit. [2019-20]



[Ans: 4 A]

Q:40. Use superposition theorem to find current I in the circuit shown in Figure below. All resistances are in ohms. [2021-22]



[Ans: 0.05 A]

Step-V: *Calculation of Load current/voltage/power:* The load's voltage, current and power may be calculated by a simple arithmetic operation only.

Load current,

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$I_L = \frac{40}{73.3 + 10}$$

$$\boldsymbol{I_L = 0.48\ A}$$

Voltage across the load may be calculated as,

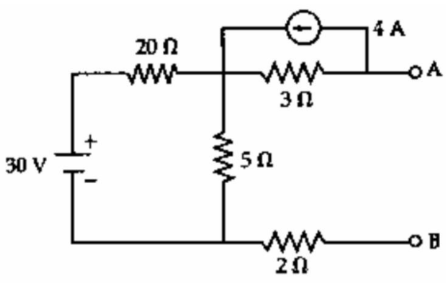
$$V_L = I_L R_L = \frac{V_{TH}}{R_{TH} + R_L} . R_L$$

☞ **Limitations of Thevenin's Theorem**

Thevenin's theorem is applicable only to linear circuits, e.g., only passive resistors, inductors, and capacitors. Thevenin's theorem is not applicable to nonlinear devices such as diodes and transistors.

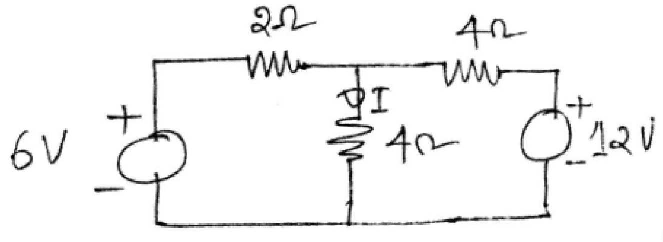
Practice Questions on Thevenin's Theorem

Q:41. Find Thevinin's equivalent circuit across A-B Shown in Figure. [2005-06]



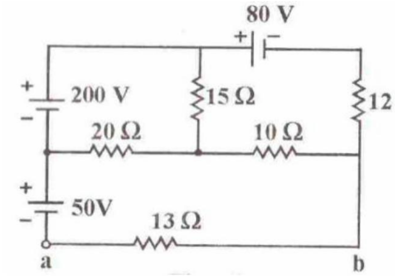
[Ans: V_{th}=-6 V, R_{th}=9Ω]

Q:42. Give the limitations of Thevenin's theorem. Find current I using this theorem in circuit in figure. [2006-07]



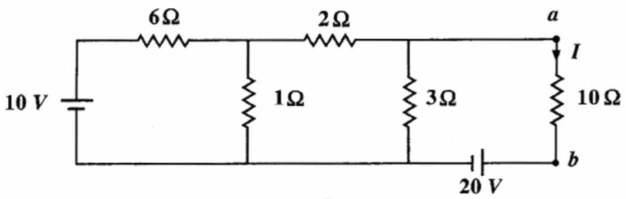
[Ans: V_{th}=8 V, R_{th}=1.33Ω, $I_{4Ω}$=1.5 A]

Q:43. Using Thevenin's theorem, determine current in 13-ohm resistance. [2007-08]



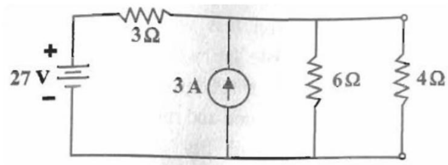
[Ans: V_{th}=167.67 V, R_{th}=7.29Ω, $I_{13Ω}$=8.26 A]

Q:44. Replace the network of the following figure to the left of terminals a-b by its Thevenin's equivalent circuit. Hence determine I. [2008-09]



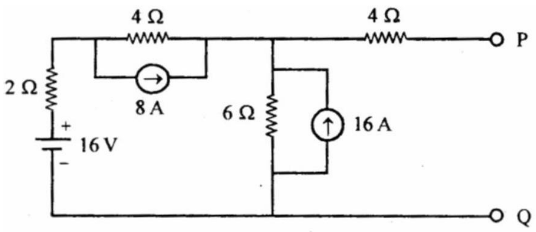
[Ans: V_{th}=-19.2 V, R_{th}=1.46Ω, $I_{4Ω}$=1.67 A]

Q:45. Determine current in 4-ohm resistance using Thevenin's theorem in the following circuit. [2009-10]



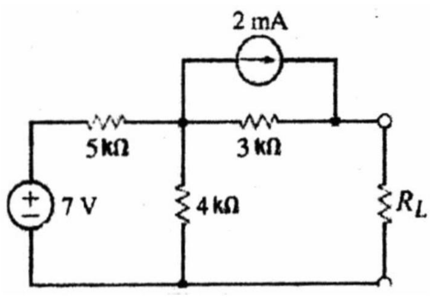
[Ans: V_{th}=24 V, R_{th}=2 Ω, $I_{4Ω}$=4 A]

Q:46. Find the V_{TH} and R_{TH} for the circuit shown in figure. [2013-14]



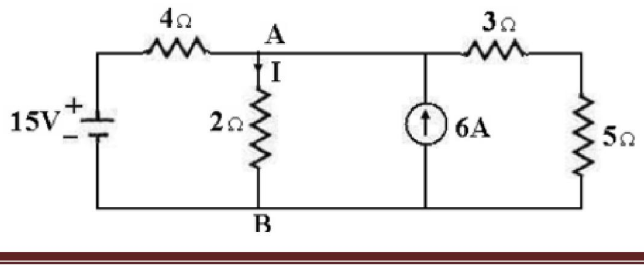
[Ans: V_{th}=72 V, R_{th}=7Ω]

Q:47. Draw the Thevenin's equivalent circuit of the figure. [2015-16]



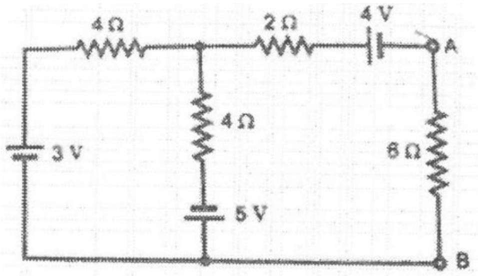
[Ans: V_{th}=9.11 V, R_{th}=5.22 kΩ]

Q:48. Determine current through 2 Ω resistor using Thevenin's theorem. [2018-19]



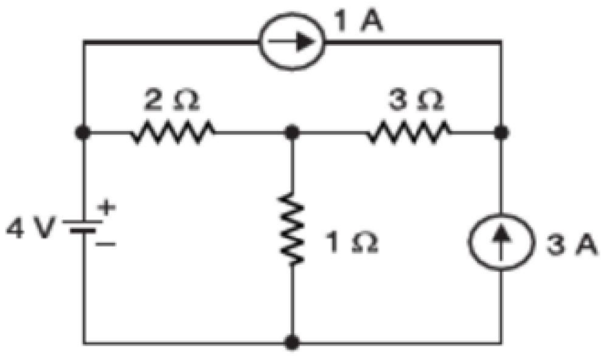
[V_{th}=26 V, R_{th}=2.67 Ω, $I_{2Ω}$=5.57 A]

Q:49. Using Thevenin's theorem, determine the current through 6 ohms. [2020-21]



[V_{th}=-5 V, R_{th}=4 Ω, $I_{6Ω}$=-0.5 A]

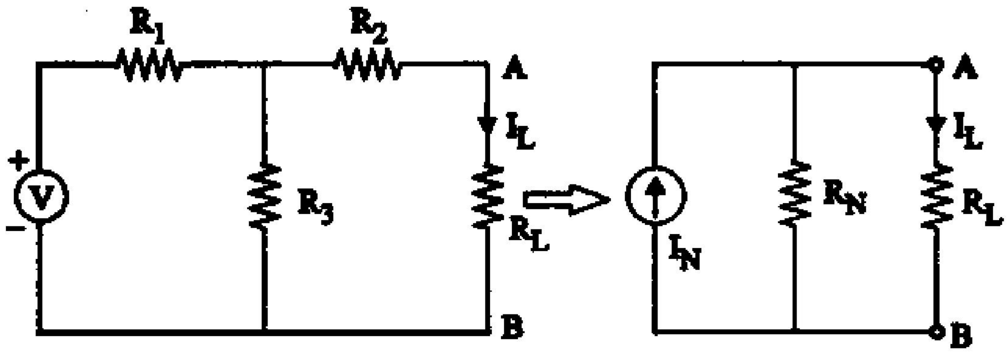
Q:50. Using Thevenin's theorem, find current in 1 Ω resistor in the circuit shown in figure below. [2021-22]



[V_{th}=6 V, R_{th}=1.2 Ω, $I_{1Ω}$=2.72 A]

9. Norton's theorem: - Norton's theorem states that

"Any two output terminals active linear network containing independent sources (includes voltage and current sources) can be replaced by a simple current source of magnitude I_N in parallel with a single resistor R_N"



Here,

☞ **I_N** - current measured in the short circuit placed across the terminal pair A & B.

☞ **R_N** - is the equivalent resistance looking into the terminal pair A & B with all independent sources has been replaced by their internal resistances.

☞ Limitations of Norton's Theorems

- It is not applicable to the circuits consisting of unilateral elements like diode etc.
- Not applicable for the circuits consisting of non-linear elements like diode, transistor etc.
- Not applicable for the circuits consisting of load series or parallel with dependent sources.
- Not applicable to the circuits consisting of magnetic coupling between load and any other circuit element.

Practice Questions on Norton's Theorem

Q:51. Determine current in 10 Ω resistance using Norton's theorem in the following network. [2005-06]

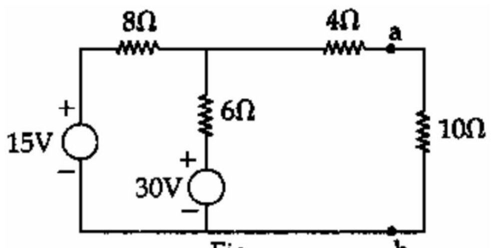
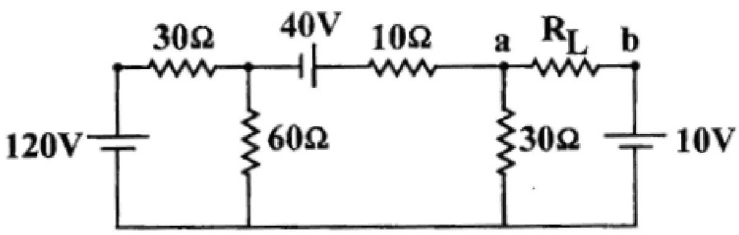


Fig.

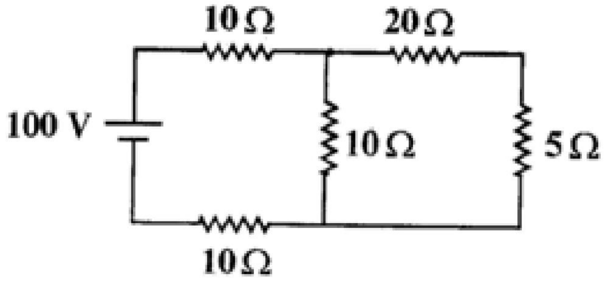
[Ans: I_N=-3.17 A, R_N=7.43 Ω, $I_{10\Omega}$=1.35 A]

Q:52. Find the Norton's equivalent circuit as seen by R_L in the following circuit. [2007-08, 2008-09]



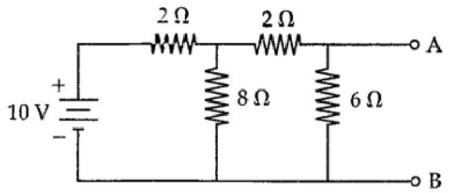
[Ans: I_N=-3.33 A, R_N=15 Ω]

Q:53. Determine the value of current through the 5Ω resistance using Norton's theorem in the circuit shown in figure. [2007-08]



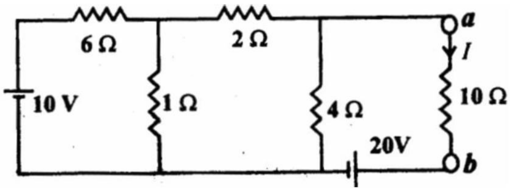
[Ans: I_N=-1.25 A, R_N=26.67 Ω, $I_{5\Omega}$=1.05 A]

Q:54. Determine Norton's equivalent across AB in figure. [2009-10]



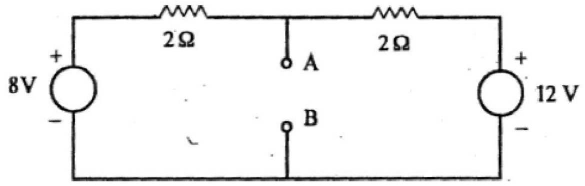
[Ans: I_N=-2.22 A, R_N=2.25 Ω]

Q:55. Convert the network at terminals ab by its Norton's equivalent circuit. Hence determine I of the circuit shown in figure. [2010-11]



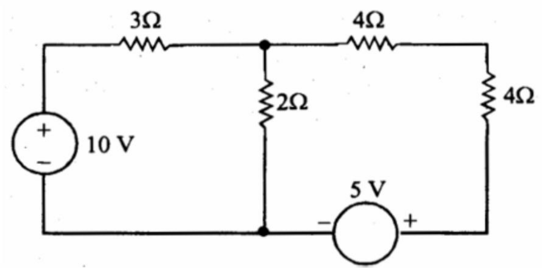
[Ans: I_N=-11.47 A, R_N=1.67 Ω, $I_{10\Omega}$=-1.64 A]

Q:56. Calculate Norton's equivalent of the network shown in figure at terminal AB. Determine the current through 4 Ω resistor across AB. [2011-12]



[Ans: I_N=10 A, R_N=1 Ω]

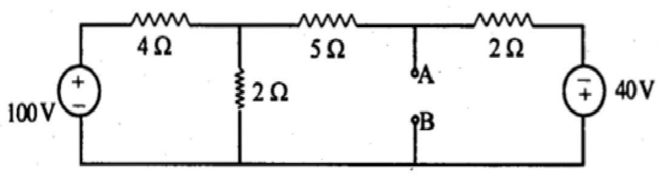
Q:57. Define Norton's theorem. Give its limitations. Find the current in 2Ω resistor using this theorem. [2012-13]



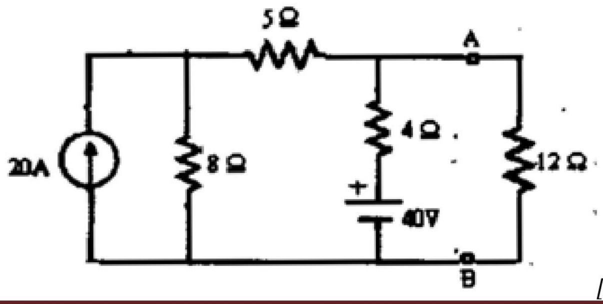
[Ans: I_N=3.96 A, R_N=2.18 Ω, $I_{2\Omega}$= 2.07A]

Q:58. Obtain the Norton's equivalent circuit at terminal AB of the network shown in figure. [2012-13]

[Ans: I_N=25.26 A, R_N=1.52 Ω]

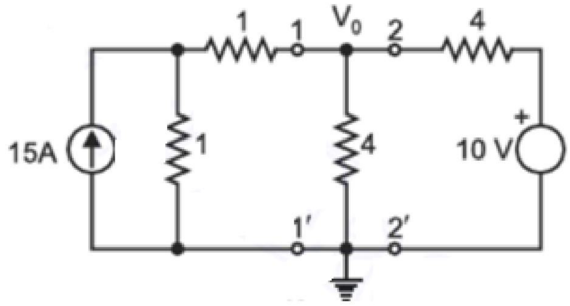


Q:59. Find the current in 12 Ω resistance using Norton's theorem for the given circuit. [2018-19]



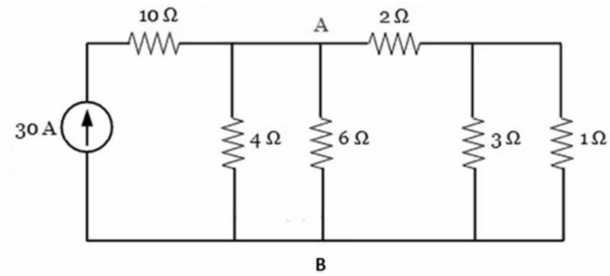
[Ans: I_N=22.30 A, R_N=3.06 Ω, $I_{12\Omega}$=4.53 A]

Q:60. With the help of Nortons theorem, find V_0 in the circuit shown below. All resistances are in ohms. [2021-22]



[Ans: I_N=10 A, R_N=1.33 Ω, $I_{4\Omega}$=2.49 A, V_o=9.96 V]

Q:61. Determine the current through A-B in the given circuit using Norton's theorem. [2021-22]



[Ans: I_N=30 A, R_N=1.62 Ω, $I_{6\Omega}$=6.37 A]

Additional Questions

Short Answer Questions (2 marks)

Q:62. Write two characteristics of Active elements. (2016-17)

Q:63. Describe the Active elements and Passive elements with examples. (2023-24, 2022-23, 2020-21, 2018-19, 2017-18, 2015-16)

Q:64. Describe briefly the Unilateral & Bilateral elements with examples. (2024-25, 2023-24, 2022-23, 2019-20, 2018-19, 2017-18, 2016-17, 2015-16)

Q:65. Define linear and non-linear elements. (2015-16)

Q:66. Define lumped and distributed elements.

Q:67. Describe the following elements briefly: (i)Independent Ideal Voltage source (ii) Independent Ideal Current source. (2017-18, 2021-22, 2022-23, 2023-24)

Q:68. Define the ideal voltage source and ideal current source. (2014-15,2017-18, 2018-19, 2020-21)

Q:69. Why is linearity important in circuits? (2021-22)

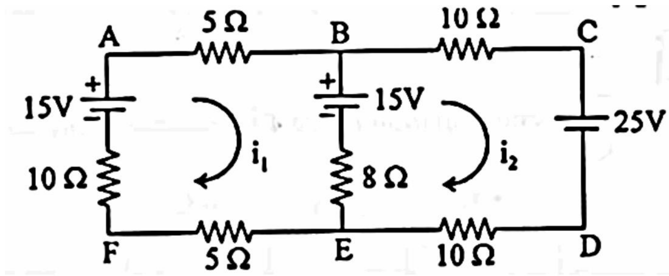
Q:70. What are Kirchoff's Laws? (2022-23, 2023-24)

Long Answer Questions

Q:71. State and explain Kirchoff's Law. What are the limitations and applications of Kirchoff's Law in circuit theory? Explain. (2016-17)

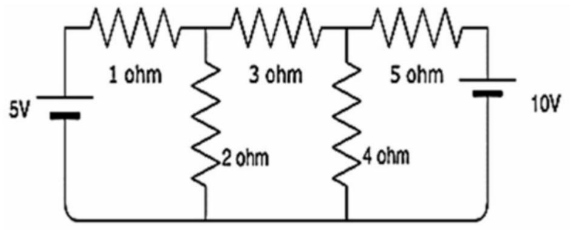
Q:72. Explain the procedure of mesh analysis with the help of an example. (2023-24)

Q:73. Find the current I_1 and I_2 using the Mesh Analysis. (2024-25)



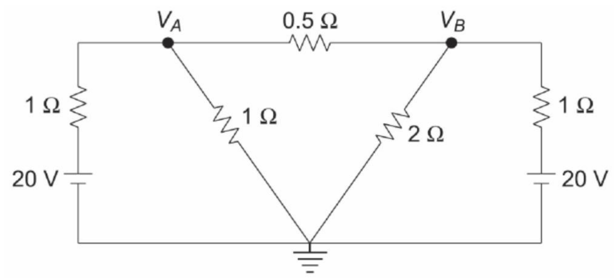
[Ans: i_1=-0.111 A, i_2=-0.389 A]

Q:74. Find the value of currents I_1, I_2 and I_3 flowing in clockwise in the first, second and third mesh respectively. (2024-25)



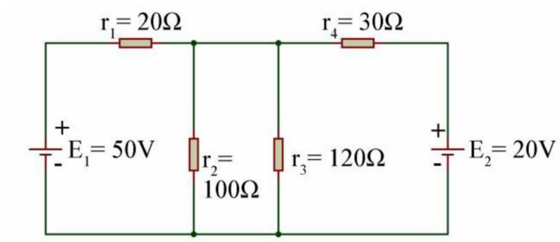
[Ans: 1.54 A, -0.189 A, -1.195 A]

Q:75. Determine the current by Nodal method, through 2 Ω resistor for the network shown below? (2022-23)



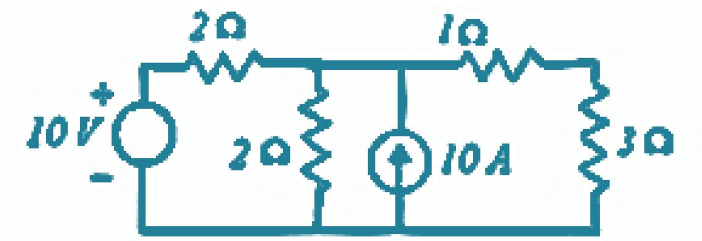
[Ans: 6 A]

Q:76. Using the Nodal method, find the current through resistor r2 in the given figure below. (2021-22)



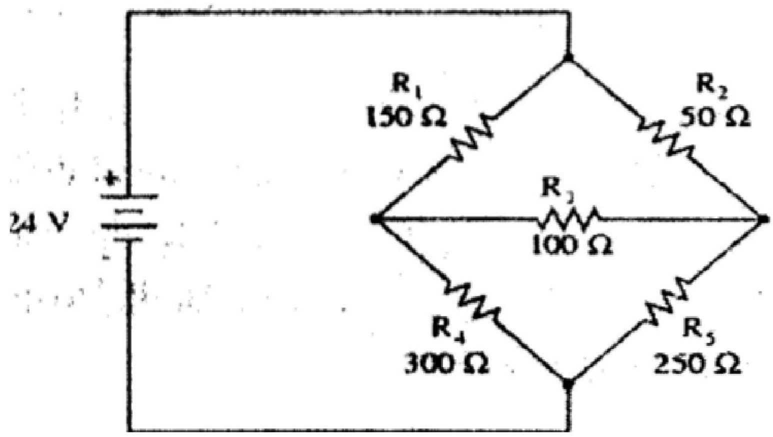
[Ans: 0.31 A]

Q:77. Find the current in 3 Ω resistance by loop current method and verify the answer by node voltage method. (2015-16)



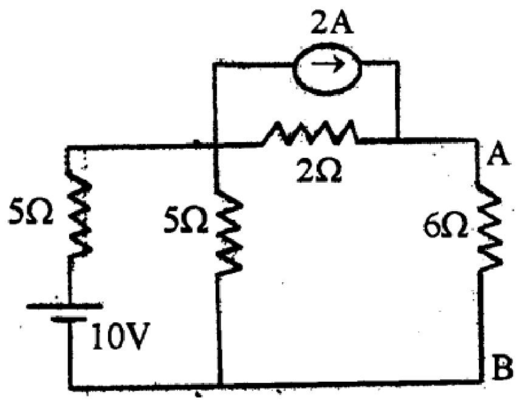
[Ans: 3 A]

Q:78. Calculate the current in R_3 by using Nodal analysis in fig.2. (2014-15)



[Ans: 16.66 mA]

Q:79. Using source transformation method to compute the current through 6 Ω resistor of following figure. (2014-15)



[Ans: 0.857 A]

Q:80. Explain voltage and current source of the network with characteristics. Explain source-transformation principle in any circuit. [2011-12]